

TABLE 2: DESIGN MATRIX

Characteristics	FR		DP	Re- lay Spe- cifi- cation	Con- trol- ler	Enclo- sure NEMA Std	Hall Effect Cur- rent Sens- er	Ambi- ent Weather Sens- or	Storage Module	Storage Spec- ifica- tion	PCB De- sign	NTC Sensor	Enclo- sure Mate- rial	En- clo- sure Col- our	Con- sole De- sign	Wi-fi Mod- ule	Brake Relay Driver	Dum- my Load Relay Driver	Dead CPU Alarm De- lay	Over- heat Ala- rm	Con- nec- tion Sta- tus Indi- cator	Org for After Sales	Org for Cus- tomer Sup- port	En- clo- sure Size	User Man- ual	Env. Norm	La- bour Prac- tice	Pow- er Sup- ply De- sign
		CODE	DP1	DP2	DP3	DP4	DP5	DP6	DP7	DP8	DP9	DP10	DP11	DP12	DP13	DP14	DP15	DP16	DP17	DP18	DP19	DP20	DP21	DP22	DP23	DP24	DP25	
Manu- facturing	Operating volt	M1	X																									
	Max current	M2	X																									
	Controller volt	M3		X																								
	Weather protection	M4			X																							
	Alternator current sense	M5		X																								
	Discharge current sense	M6		X																								
	Batt. voltage sense	M7		X																								
	Batt. current sense	M8				X																						
	Alternator voltage sense	M9		X																								
	Temp sense	M10					X																					
	Humidity sense	M11					X																					
	Data storage technology	M12						X	X																			
	Data storage size	M13								X																		
	Isolate connecting terminals from main controller	M14									X																	
	Sense temp of terminal block	M15										X																
Materials	Provide heat and fire resistant material	T1										X																
	Box colour	T2											X															
Software	User Interface	S1												X														
	Connectivity	S2													X													
	If battery absent, connect turbine to dummy load	S3														X												
	Activate break circuit when dummy load is missing	S4														X												
	If battery shorted, connect power to dummy load	S5															X											
	If dummy load is shorted then activate brake	S6														X												
	When wind speed exceeds a set threshold activate brake	S7														X												
	By default connect turbine to dummy load	S8																X										
	Test software functioning at regular interval	S9																	X									
	Alarm if terminals over heat	S10										X								X								
	Raise warning from server/controller if connection fails	S11																			X							
Organis- ation	After sales service	G1																				X						
	Help line	G2																					X					
	User manual	G3																							X			
Systems	Device size	T1																						X				
	Safety margin	T2	X																									
	Parts environmental Norm	T3																							X			
	Sustainability norm	T4																								X	X	
	Alternate power source for circuit	T5																									X	
	Check CPU function and sound alarm if abnormal condition detected	T6																	X									
Business	Labour norm	B1																										